



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

**FINAL EXAMINATION
JANUARY 2024**

Course: EC3123/EC3239 (Operating Systems Principles)

Time: 2.00PM – 5.00PM
(3 hours)

Lecturer: Mohd Shahdi Bin Ahmad

Date: 13 May 2024

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **2** printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

- 1 a. Identify **THREE (3)** types of Dynamic Memory Allocation. **(6 MARKS)**
- b. Illustrates a single and multiple partitions in memory for contiguous memory allocation. **(10 MARKS)**
- c. Illustrates a Memory Compaction in a block diagram **(9 MARKS)**

- 2 a. Identify the steps involved in SRTF (Shortest Remaining Time First) scheduling. **(8 MARKS)**
- b. Based on Table 1, answers the following questions:

Process	Arrival Time	Burst Time
P1	0	3
P2	1	5
P3	2	2

Table 1

- i. Draw the Gantt chart for P1, P2 and P3. **(11 MARKS)**
 - ii. Based on the Gantt chart from question 2b(i), write the average turnaround time and average waiting time. **(6 MARKS)**
- 3 a. Explain deadlock in computer systems and how it happens. **(4 MARKS)**
 - b. Describe the **FOUR (4)** necessary conditions for a deadlock to occur in a system. **(12 MARKS)**
 - c. Describe the **THREE (3)** strategies to handle the deadlock. **(9 MARKS)**
- 4 a. Write a Linux command based on the instructions below:
 - i. Create a folder named "University." **(2 MARKS)**
 - ii. Remove the non-empty folder named "Student". **(2 MARKS)**

b. Based on Table 2, answer the following questions:

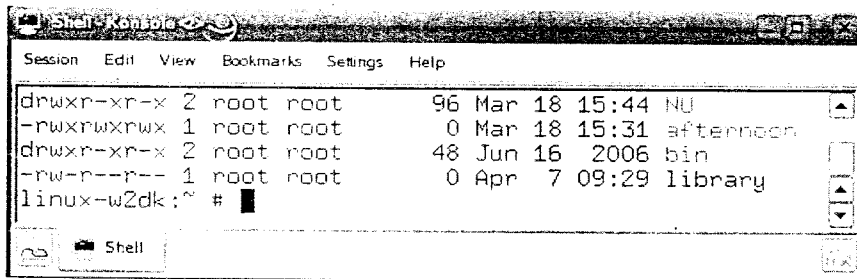
No.	Group Name	Group ID	Username	User ID
1	UIUX	3000	David	3001
2	Programmer	4000	Lim	4001
3	Marketing	5000	Husin	5001
4	Marketing	5000	Edry	5002

Table 2

i. Show Linux commands for creating all the groups. (6 MARKS)

ii. Show Linux commands for creating all the usernames. (8 MARKS)

c.



```

Shell - Konsole
Session Edit View Bookmarks Settings Help
drwxr-xr-x 2 root root 96 Mar 18 15:44 NU
-rwxrwxrwx 1 root root 0 Mar 18 15:31 afternoon
drwxr-xr-x 2 root root 48 Jun 16 2006 bin
-rw-r--r-- 1 root root 0 Apr 7 09:29 library
linux-w2dk:~ #
  
```

Figure 1

Figure 1 shows the permission of a file named "library". Provide the command based on the instructions given:

i. Identify the permission for the file named "library". (1 MARK)

ii. Modify the file permissions so that all users and groups can read and write. (3 MARKS)

iii. Modify the owner of the file to John. (3 MARKS)

-END OF QUESTION PAPER-